```
OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;

/*  STAT 382 SAS HOMEWORK 1 */



/* Task 1: Data Frame Creation */
TITLE1 "Task 1";
TITLE2 "Q1 Part a) Import Datasets";
PROC IMPORT datafile="/home/u64394289/SAS_cars_domestic.csv"
OUT=domestic
DBMS=CSV
replace;
run;

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to
WORK.PARMS.PARMS.SLIST.
 /**********************************************************************
 *   PRODUCT:   SAS
 *   VERSION:   9.4
 *   CREATOR:   External File Interface
 *   DATE:      25NOV25
 *   DESC:      Generated SAS Datastep Code
 *   TEMPLATE SOURCE:  (None Specified.)
 ***********************************************************************/
    data WORK.DOMESTIC    ;
    %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
    infile '/home/u64394289/SAS_cars_domestic.csv' delimiter = ',' MISSOVER DSD lrecl=32767 firstobs=2 ;
       informat Name $18. ;
       informat Car_Type $9. ;
       informat Drive_Wheels_Available $12. ;
       informat Seating best32. ;
       informat Engine_liter best32. ;
       informat HP best32. ;
       informat Length_in best32. ;
       informat Width_in best32. ;
       informat Height_in best32. ;
       informat Wheelbase_in best32. ;
       informat Weight_lbs best32. ;
       informat Cargo_Volume_cuft best32. ;
       informat Overall_MPG best32. ;
       informat Lowest_MSRP best32. ;
       informat Parent_Company_HQ $3. ;
       informat Manufacturer_Type $8. ;
       format Name $18. ;
       format Car_Type $9. ;
       format Drive_Wheels_Available $12. ;
       format Seating best12. ;
       format Engine_liter best12. ;
       format HP best12. ;
       format Length_in best12. ;
       format Width_in best12. ;
       format Height_in best12. ;
       format Wheelbase_in best12. ;
       format Weight_lbs best12. ;
       format Cargo_Volume_cuft best12. ;
       format Overall_MPG best12. ;
       format Lowest_MSRP best12. ;
       format Parent_Company_HQ $3. ;
       format Manufacturer_Type $8. ;
    input
                Name  $
                Car_Type  $
                Drive_Wheels_Available  $
                Seating
                Engine_liter
                HP
                Length_in
                Width_in
                Height_in
                Wheelbase_in
                Weight_lbs
                Cargo_Volume_cuft
                Overall_MPG
                Lowest_MSRP
                Parent_Company_HQ  $
                Manufacturer_Type  $
    ;
```

```
          if _ERROR_ then call symputx('_EFIERR_',1);  /* set ERROR detection macro variable */
          run;

NOTE: The infile '/home/u64394289/SAS_cars_domestic.csv' is:
      Filename=/home/u64394289/SAS_cars_domestic.csv,
      Owner Name=u64394289,Group Name=oda,
      Access Permission=-rw-r--r--,
      Last Modified=24Nov2025:20:46:47,
      File Size (bytes)=5086

NOTE: 57 records were read from the infile '/home/u64394289/SAS_cars_domestic.csv'.
      The minimum record length was 73.
      The maximum record length was 92.
NOTE: The data set WORK.DOMESTIC has 57 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              9505.43k
      OS Memory           32796.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        192  Switch Count  2
      Page Faults                       0
      Page Reclaims                     172
      Page Swaps                        0
      Voluntary Context Switches        14
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


57 rows created in WORK.DOMESTIC from /home/u64394289/SAS_cars_domestic.csv.



NOTE: WORK.DOMESTIC data set was successfully created.
NOTE: The data set WORK.DOMESTIC has 57 observations and 16 variables.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.05 seconds
      user cpu time       0.03 seconds
      system cpu time     0.00 seconds
      memory              9505.43k
      OS Memory           33056.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        192  Switch Count  10
      Page Faults                       0
      Page Reclaims                     1342
      Page Swaps                        0
      Voluntary Context Switches        87
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           328



      PROC IMPORT datafile="/home/u64394289/SAS_cars_foreign.csv"
      OUT=foreign
      DBMS=CSV
      replace;
      run;

NOTE: Unable to open parameter catalog: SASUSER.PARMS.PARMS.SLIST in update mode. Temporary parameter values will be saved to
WORK.PARMS.PARMS.SLIST.
       /**********************************************************************
       *   PRODUCT:   SAS
       *   VERSION:   9.4
       *   CREATOR:   External File Interface
       *   DATE:      25NOV25
       *   DESC:      Generated SAS Datastep Code
       *   TEMPLATE SOURCE:  (None Specified.)
       ***********************************************************************/
          data WORK.FOREIGN    ;
          %let _EFIERR_ = 0; /* set the ERROR detection macro variable */
          infile '/home/u64394289/SAS_cars_foreign.csv' delimiter = ',' MISSOVER DSD lrecl=32767 firstobs=2 ;
             informat Name $18. ;
             informat Car_Type $11. ;
             informat Drive_Wheels_Available $12. ;
             informat Seating best32. ;
             informat Engine_liter best32. ;
             informat HP best32. ;
```

```
              informat Length_in best32. ;
              informat Width_in best32. ;
              informat Height_in best32. ;
              informat Wheelbase_in best32. ;
              informat Weight_lbs best32. ;
              informat Cargo_Volume_cuft best32. ;
              informat Overall_MPG best32. ;
              informat Lowest_MSRP best32. ;
              informat Parent_Company_HQ $7. ;
              informat Manufacturer_Type $7. ;
              format Name $18. ;
              format Car_Type $11. ;
              format Drive_Wheels_Available $12. ;
              format Seating best12. ;
              format Engine_liter best12. ;
              format HP best12. ;
              format Length_in best12. ;
              format Width_in best12. ;
              format Height_in best12. ;
              format Wheelbase_in best12. ;
              format Weight_lbs best12. ;
              format Cargo_Volume_cuft best12. ;
              format Overall_MPG best12. ;
              format Lowest_MSRP best12. ;
              format Parent_Company_HQ $7. ;
              format Manufacturer_Type $7. ;
           input
                       Name  $
                       Car_Type  $
                       Drive_Wheels_Available  $
                       Seating
                       Engine_liter
                       HP
                       Length_in
                       Width_in
                       Height_in
                       Wheelbase_in
                       Weight_lbs
                       Cargo_Volume_cuft
                       Overall_MPG
                       Lowest_MSRP
                       Parent_Company_HQ  $
                       Manufacturer_Type  $
           ;
           if _ERROR_ then call symputx('_EFIERR_',1);  /* set ERROR detection macro variable */
           run;

NOTE: The infile '/home/u64394289/SAS_cars_foreign.csv' is:
      Filename=/home/u64394289/SAS_cars_foreign.csv,
      Owner Name=u64394289,Group Name=oda,
      Access Permission=-rw-r--r--,
      Last Modified=24Nov2025:20:46:47,
      File Size (bytes)=14083

NOTE: 165 records were read from the infile '/home/u64394289/SAS_cars_foreign.csv'.
      The minimum record length was 69.
      The maximum record length was 98.
NOTE: The data set WORK.FOREIGN has 165 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              9503.06k
      OS Memory           32796.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        193  Switch Count  2
      Page Faults                       0
      Page Reclaims                     159
      Page Swaps                        0
      Voluntary Context Switches        14
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           272


165 rows created in WORK.FOREIGN from /home/u64394289/SAS_cars_foreign.csv.



NOTE: WORK.FOREIGN data set was successfully created.
```

```
NOTE: The data set WORK.FOREIGN has 165 observations and 16 variables.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.03 seconds
      user cpu time       0.03 seconds
      system cpu time     0.01 seconds
      memory              9503.06k
      OS Memory           33056.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        193  Switch Count  10
      Page Faults                       0
      Page Reclaims                     1137
      Page Swaps                        0
      Voluntary Context Switches        82
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           288


           TITLE2 "Q1 Part b) Modify column formats";
           DATA domestic; *change the name to whatever you named your DOMESTIC dataset;
             LENGTH Manufacturer_Type $30 Parent_Company_HQ $30 Car_Type $14;
             FORMAT Manufacturer_Type $30. Parent_Company_HQ $30. Car_Type $14.;
             SET domestic; *change the name to whatever you named your DOMESTIC dataset;
           RUN;

NOTE: There were 57 observations read from the data set WORK.DOMESTIC.
NOTE: The data set WORK.DOMESTIC has 57 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1022.50k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        194  Switch Count  2
      Page Faults                       0
      Page Reclaims                     119
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


           DATA foreign; *change the name to whatever you named your FOREIGN dataset;
             LENGTH Manufacturer_Type $30 Parent_Company_HQ $30 Car_Type $14;
             FORMAT Manufacturer_Type $30. Parent_Company_HQ $30. Car_Type $14.;
             SET foreign;  *change the name to whatever you named your FOREIGN dataset;
           RUN;

NOTE: There were 165 observations read from the data set WORK.FOREIGN.
NOTE: The data set WORK.FOREIGN has 165 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              959.62k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        195  Switch Count  2
      Page Faults                       0
      Page Reclaims                     114
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           272


           TITLE2 "Q1 Part c) Combine DOMESTIC and FOREIGN Datasets"; * list DOMESTIC first;
           data all_cars;
           set domestic foreign;
           run;

NOTE: There were 57 observations read from the data set WORK.DOMESTIC.
```

```
NOTE: There were 165 observations read from the data set WORK.FOREIGN.
NOTE: The data set WORK.ALL_CARS has 222 observations and 16 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1357.40k
      OS Memory           28080.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        196  Switch Count  2
      Page Faults                       0
      Page Reclaims                     145
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264



     TITLE2 "Q1 Part d) Sort all_cars by ascending Drive_Wheels_Available";
     proc sort data = all_cars;
     by Drive_Wheels_Available;
     run;

NOTE: There were 222 observations read from the data set WORK.ALL_CARS.
NOTE: The data set WORK.ALL_CARS has 222 observations and 16 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              926.31k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        197  Switch Count  2
      Page Faults                       0
      Page Reclaims                     103
      Page Swaps                        0
      Voluntary Context Switches        13
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264



     TITLE2 "Q1 Part e) Create the dataset DriveWheels";
     data DriveWheels;
     length Drive_Wheels_Available $12 Drive_Wheels_Max $10;
     input Drive_Wheels_Available $1-12 Drive_Wheels_Max $14-23;
     datalines;

NOTE: The data set WORK.DRIVEWHEELS has 9 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              739.84k
      OS Memory           27560.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        198  Switch Count  2
      Page Faults                       0
      Page Reclaims                     87
      Page Swaps                        0
      Voluntary Context Switches        12
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


     ;
     run;

     proc print data = DriveWheels;
     run;

NOTE: There were 9 observations read from the data set WORK.DRIVEWHEELS.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.01 seconds
```

```
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              1375.96k
      OS Memory           27560.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        199  Switch Count  0
      Page Faults                       0
      Page Reclaims                     62
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           8


      proc contents data = DriveWheels;
      run;

NOTE: PROCEDURE CONTENTS used (Total process time):
      real time           0.02 seconds
      user cpu time       0.02 seconds
      system cpu time     0.01 seconds
      memory              1334.37k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        200  Switch Count  0
      Page Faults                       0
      Page Reclaims                     130
      Page Swaps                        0
      Voluntary Context Switches        3
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           8


      TITLE2 "Q1 Part f) Sort DriveWheels by ascending Drive_Wheels_Available";
      proc sort data = DriveWheels;
      by Drive_Wheels_Available;
      run;

NOTE: There were 9 observations read from the data set WORK.DRIVEWHEELS.
NOTE: The data set WORK.DRIVEWHEELS has 9 observations and 2 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              932.87k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        201  Switch Count  2
      Page Faults                       0
      Page Reclaims                     114
      Page Swaps                        0
      Voluntary Context Switches        12
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


      TITLE2 "Q1 Part g) Merge the all_cars dataset and the DriveWheels dataset to create all_cars2";
      data all_cars2;
      merge all_cars DriveWheels;
      by Drive_Wheels_Available;
      run;

NOTE: There were 222 observations read from the data set WORK.ALL_CARS.
NOTE: There were 9 observations read from the data set WORK.DRIVEWHEELS.
NOTE: The data set WORK.ALL_CARS2 has 222 observations and 17 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1405.62k
      OS Memory           28080.00k
      Timestamp           11/25/2025 03:58:24 AM
```

```
      Step Count                        202  Switch Count  2
      Page Faults                       0
      Page Reclaims                     144
      Page Swaps                        0
      Voluntary Context Switches        14
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264
```

```
/* Task 2: Cleaning and Formatting */
TITLE1 "Task 2";
TITLE2 "Q2) Replace missing values for Engine_liter with 0s in new dataset cars_fix";
data cars_fix;
set all_cars2;
if Engine_liter = . then Engine_liter = 0;
run;
```

```
NOTE: There were 222 observations read from the data set WORK.ALL_CARS2.
NOTE: The data set WORK.CARS_FIX has 222 observations and 17 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              959.12k
      OS Memory           27820.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        203  Switch Count  2
      Page Faults                       0
      Page Reclaims                     110
      Page Swaps                        0
      Voluntary Context Switches        13
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264
```

```
TITLE2 "Q3) Remove rows for Car_Type and Overall_MPG that are missing. Do this in a new dataset called cars_clean";
data cars_clean;
set cars_fix;
if Car_Type = "" then delete;
if Overall_MPG = . then delete;
run;
```

```
NOTE: There were 222 observations read from the data set WORK.CARS_FIX.
NOTE: The data set WORK.CARS_CLEAN has 216 observations and 17 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1020.53k
      OS Memory           28076.00k
      Timestamp           11/25/2025 03:58:24 AM
      Step Count                        204  Switch Count  2
      Page Faults                       0
      Page Reclaims                     113
      Page Swaps                        0
      Voluntary Context Switches        10
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           264
```

```
306
307
308        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
318
```